Australian Election Outcomes

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This is my abstract

My first PDF Submission

Here I will provide a PDF on Australian Election Outcomes

This analysis was conducted using R (R Core Team 2021) and follows the workflow described by Alexander (Alexander 2023)

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'readr' was built under R version 4.4.3

Warning: package 'purrr' was built under R version 4.4.3

Warning: package 'dplyr' was built under R version 4.4.3

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.5.1
v ggplot2   4.0.3     v tibble    3.2.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
simulated_data <-
  tibble(
    # Use 1 through to 151 to represent each division
    "Division" = 1:151,
    # Randomly pick an option, with replacement, 151 times
    "Party" = sample(
      x = c("Liberal", "Labor", "National", "Green", "Other"),
      size = 151,
      replace = TRUE
    )
  )

simulated_data
```

```
# A tibble: 151 x 2
   Division Party
      <int> <chr>
 1        1 National
 2        2 Liberal
 3        3 Liberal
 4        4 Labor
 5        5 National
 6        6 Other
 7        7 Green
 8        8 Other
 9        9 Labor
10       10 National
# i 141 more rows
```

```
#### Read in the data ####
raw_elections_data <-
  read_csv(
```

```
    file =
      "https://results.aec.gov.au/27966/website/Downloads/HouseMembersElectedDownload-27966.c
    show_col_types = FALSE,
    skip = 1
  )

# We have read the data from the AEC website. We may like to save
# it in case something happens or they move it.
write_csv(
  x = raw_elections_data,
  file = "australian_voting.csv"
)
```

```
head(raw_elections_data)
```

```
# A tibble: 6 x 8
  DivisionID DivisionNm StateAb CandidateID GivenNm   Surname   PartyNm  PartyAb
       <dbl> <chr>      <chr>         <dbl> <chr>     <chr>     <chr>    <chr>
1        179 Adelaide   SA            36973 Steve     GEORGANAS Austral~ ALP
2        197 Aston      VIC           36704 Alan      TUDGE     Liberal  LP
3        198 Ballarat   VIC           36409 Catherine KING      Austral~ ALP
4        103 Banks      NSW           37018 David     COLEMAN   Liberal  LP
5        180 Barker     SA            37083 Tony      PASIN     Liberal  LP
6        104 Barton     NSW           36820 Linda     BURNEY    Austral~ ALP
```

```
tail(raw_elections_data)
```

```
# A tibble: 6 x 8
  DivisionID DivisionNm StateAb CandidateID GivenNm    Surname  PartyNm  PartyAb
       <dbl> <chr>      <chr>         <dbl> <chr>      <chr>    <chr>    <chr>
1        152 Wentworth  NSW           37451 Allegra    SPENDER  Indepen~ IND
2        153 Werriwa    NSW           36810 Anne Maree STANLEY  Austral~ ALP
3        150 Whitlam    NSW           36811 Stephen    JONES    Austral~ ALP
4        178 Wide Bay   QLD           37506 Llew       O'BRIEN  Liberal~ LNP
5        234 Wills      VIC           36452 Peter      KHALIL   Austral~ ALP
6        316 Wright     QLD           37500 Scott      BUCHHOLZ Liberal~ LNP
```

```
#### Basic cleaning ####
raw_elections_data <-
  read_csv(
```

```
    file = "australian_voting.csv",
    show_col_types = FALSE
  )
```

```
# Make the names easier to type
cleaned_elections_data <-
  clean_names(raw_elections_data)

# Have a look at the first six rows
head(cleaned_elections_data)
```

```
# A tibble: 6 x 8
  division_id division_nm state_ab candidate_id given_nm  surname   party_nm
        <dbl> <chr>       <chr>           <dbl> <chr>     <chr>     <chr>
1         179 Adelaide    SA              36973 Steve     GEORGANAS Australian ~
2         197 Aston       VIC             36704 Alan      TUDGE     Liberal
3         198 Ballarat    VIC             36409 Catherine KING      Australian ~
4         103 Banks       NSW             37018 David     COLEMAN   Liberal
5         180 Barker      SA              37083 Tony      PASIN     Liberal
6         104 Barton      NSW             36820 Linda     BURNEY    Australian ~
# i 1 more variable: party_ab <chr>
```

```
cleaned_elections_data <-
  cleaned_elections_data |>
  select(
    division_nm,
    party_nm
  )

head(cleaned_elections_data)
```

```
# A tibble: 6 x 2
  division_nm party_nm
  <chr>       <chr>
1 Adelaide    Australian Labor Party
2 Aston       Liberal
3 Ballarat    Australian Labor Party
4 Banks       Liberal
5 Barker      Liberal
6 Barton      Australian Labor Party
```

```
names(cleaned_elections_data)
```

```
[1] "division_nm" "party_nm"
```

```
cleaned_elections_data <-
  cleaned_elections_data |>
  rename(
    division = division_nm,
    elected_party = party_nm
  )

head(cleaned_elections_data)
```

```
# A tibble: 6 x 2
  division elected_party
  <chr>    <chr>
1 Adelaide Australian Labor Party
2 Aston    Liberal
3 Ballarat Australian Labor Party
4 Banks    Liberal
5 Barker   Liberal
6 Barton   Australian Labor Party
```

```
cleaned_elections_data$elected_party |>
  unique()
```

```
[1] "Australian Labor Party"
[2] "Liberal"
[3] "Liberal National Party of Queensland"
[4] "The Greens"
[5] "The Nationals"
[6] "Independent"
[7] "Katter's Australian Party (KAP)"
[8] "Centre Alliance"
```

```
cleaned_elections_data <-
  cleaned_elections_data |>
  mutate(
    elected_party =
      case_match(
```

```
        elected_party,
        "Australian Labor Party" ~ "Labor",
        "Liberal National Party of Queensland" ~ "Liberal",
        "Liberal" ~ "Liberal",
        "The Nationals" ~ "Nationals",
        "The Greens" ~ "Greens",
        "Independent" ~ "Other",
        "Katter's Australian Party (KAP)" ~ "Other",
        "Centre Alliance" ~ "Other"
      )
  )
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `elected_party = case_match(...)`.
Caused by warning:
! `case_match()` was deprecated in dplyr 1.2.0.
i Please use `recode_values()` instead.
```

```
head(cleaned_elections_data)
```

```
# A tibble: 6 x 2
  division elected_party
  <chr>    <chr>
1 Adelaide Labor
2 Aston    Liberal
3 Ballarat Labor
4 Banks    Liberal
5 Barker   Liberal
6 Barton   Labor
```

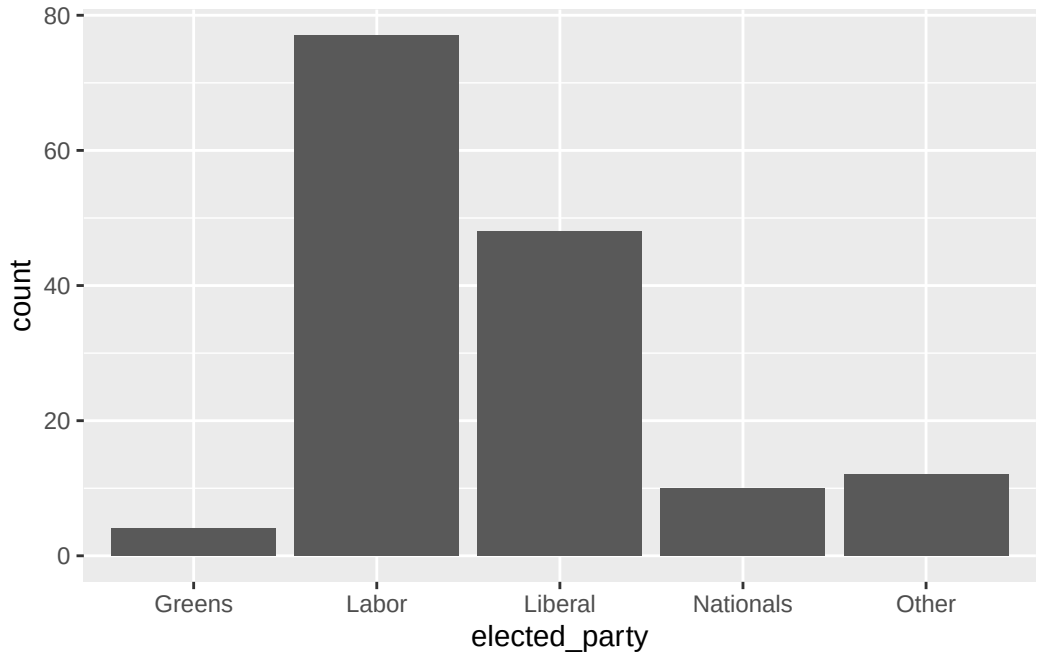
```
write_csv(
  x = cleaned_elections_data,
  file = "cleaned_elections_data.csv"
)
```

```
#### Read in the data ####
cleaned_elections_data <-
  read_csv(
    file = "cleaned_elections_data.csv",
    show_col_types = FALSE
  )
```

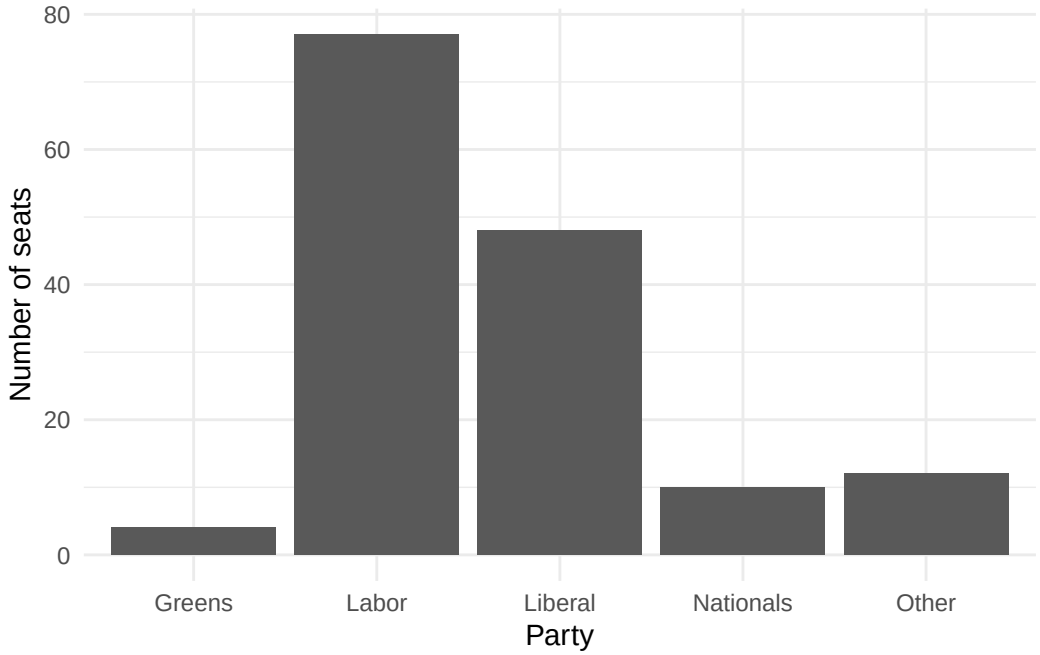
```
cleaned_elections_data |>
  count(elected_party)
```

```
# A tibble: 5 x 2
  elected_party     n
  <chr>         <int>
1 Greens            4
2 Labor            77
3 Liberal          48
4 Nationals        10
5 Other            12
```

```
cleaned_elections_data |>
  ggplot(aes(x = elected_party)) + # aes abbreviates "aesthetics"
  geom_bar()
```




```
cleaned_elections_data |>
  ggplot(aes(x = elected_party)) +
  geom_bar() +
  theme_minimal() + # Make the theme neater
  labs(x = "Party", y = "Number of seats") # Make labels more meaningful
```



References

Alexander, Rohan. 2023. *Telling Stories with Data*. Chapman; Hall/CRC. https://tellingstorieswithdata.com.

R Core Team. 2021. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.